

Project Proposal: Automated Data Pipeline for Sales Analytics

Executive Summary

We propose building a **real-time data pipeline** that ingests sales transactions from multiple sources, transforms them into a unified schema, and loads the results into a cloud analytics platform^[1^]. This will enable *executives* to generate up-to-date dashboards and support data-driven decisions.

"Data is the new oil." — John Doe, 2023'

Objectives

- **Collect** transaction logs from POS, e-commerce, and CRM systems.
- **Normalize** fields (date format, currency, product codes).
- **Store** cleansed data in a columnar database for fast querying.

- **Visualize** key metrics through scheduled reports.

Tasks

1. Set up Kafka topics for each source system.
2. Develop Spark jobs to perform ETL and quality checks.
3. Deploy Airflow DAGs for orchestration^[2^].
4. Configure Snowflake tables and access roles.
5. Create Tableau dashboards and schedule refreshes.

Technical Stack

```
# Example Docker Compose for local development
version: '3'

services:
  kafka:
    image:
confluentinc/cp-kafka:latest
    ports:
      - "9092:9092"
  spark:
    image:
bitnami/spark:latest
    depends_on:
      - kafka
```

Risks & Mitigations

- **Data latency:** use incremental pulls and windowed joins.

- **Schema drift:** implement automated schema registry checks.

Phase	Duration
Design	2 weeks
Development	6 weeks
Testing	3 weeks
Deployment	1 week

Budget Estimate

Total cost: \$85,000 covering personnel, cloud services, and tooling.

References

[1] Smith, A. (2022). *Real-time Data Integration*. Journal of Big Data, 5(3), 112–

Timeline

130.
[2] Johnson, L. (2023). *Airflow for Enterprise Workflows*. TechCrunch, 7(4), 45–59.

Footnotes

¹ *John Doe* is a fictional data analyst referenced in industry publications.
² Airflow provides scheduling and monitoring capabilities for ETL jobs.